

Credit Risk Intelligence Platform

Day 1: Project Charter and Product Requirements Document

1. The Problem Statement

When a person or a business applies for a loan, banks and lending companies currently spend hours searching through messy financial paperwork, bank statements, and tax forms to determine if the applicant can safely pay back the money. This manual review process is incredibly slow, highly repetitive, and leaves borrowers completely in the dark regarding the status of their application.

Standard software systems only look backward at past data. They simply stamp applications as "Approved" or "Rejected" without providing any explanation as to why. If an applicant's profile is even slightly weak, the borrower is turned away without receiving any guidance on how to fix their financial situation.

Project Scope and Boundaries

- **Target Users and Context:** The platform is designed for credit analysts working inside commercial lending banks. It is also built for loan applicants who want to track their active loan applications.
- **The Core Pain Point:** Extracting risk data manually from documents is slow and prone to errors. Furthermore, traditional rejection processes lack transparency and leave borrowers without a path to financial recovery.
- **The Proposed Solution:** We are creating an intelligent assistant that acts as a dynamic advisory engine. This tool will extract data, chart growth trends, provide personalized alternative loan options, and offer specific roadmaps to help users fix their credit.
- **Measures of Success:** The project will be considered successful if it reduces the total document evaluation time from several hours down to just minutes. It should also increase customer conversion rates by offering secondary, backup loan options.
- **Data Sources:** The system will process official financial PDF documents uploaded directly by the user. These include Bank Statements, Income Tax Returns (ITR), Balance Sheets, and Profit & Loss statements.
- **Out of Scope:** The system will not automatically approve or reject loans without human sign-off. It will also not handle the actual infrastructure for transferring funds.

2. Project Type and Capabilities

Feature Category	Priority Level	AI Role and Technical Implementation
------------------	----------------	--------------------------------------

Document Summarizer	Primary	Uses text chunking (split-and-refine) to process raw uploaded files and create a clean, visual overview of the user's finances.
AI Chat Assistant	Secondary	Provides conversational follow-up questions and answers. The chat is strictly based on the uploaded document contents to help resolve any questions the credit analyst might have.
Recommendation System	Primary	Uses intelligent business rules to suggest personalized ways to fix credit issues and recommends safer, alternative loan structures.

3. Main Features and User Interface Plan

Core Recommendation Features

- **Actionable Credit Checklist:** If a borrower's application is rejected, the system lists simple, step-by-step actions to fix their profile. Examples include suggesting they pay off a specific credit card or reduce their overall card utilization so they can get approved the next time.
- **Safer Backup Loan Options:** If the requested loan amount carries too much risk, the system automatically calculates a safer alternative offer. For example, it might recommend a \$30,000 loan instead of a \$50,000 loan, or it might ask the applicant to provide a business asset as security.
- **Long-Term Risk Simulator:** The AI looks ahead to model future cash flows and potential economic changes, such as higher interest rates. This ensures the borrower can comfortably afford their monthly payments in the upcoming year.
- **Urgency Pipeline Sorting:** The AI automatically scans through the text in applications to identify applicants who need funds immediately, such as for an emergency company payroll. This allows bank teams to handle high-priority leads first.

Document Analysis Features

- **Borrower Loan Pros & Cons:** This feature shows applicants exactly where their financial profile is strong, like having a stable income. It also highlights weaknesses, such as having too much existing debt, ensuring they understand their exact chances of approval.
- **Financial PDF Uploads:** Users can upload the official files required for their evaluation. Supported documents include Bank Statements, Income Tax Returns (ITR), Audited Balance Sheets, and Company Profit & Loss statements.
- **Visual Dashboard Summaries:** The system creates an interactive screen featuring a 3-year profit growth graph and a loan-reduction tracker chart. It also provides immediate red-flag alerts if the applicant missed past loan payments.
- **Pre-configured Clickable Prompts:** The platform provides ready-to-use, clickable sample questions tailored to different user types:
 - **For Banks:** Prompts include "Did this company miss any payments in the last 12 months?" or "Does the borrower have enough cash for next month's installment?"
 - **For Borrowers:** Prompts include "What is causing my credit score to drop?" or "Based on my documents, how much loan amount can I safely afford?"

User Interface (UI) Layout Plan

- **Left Sidebar (File Upload Area):** This area contains clean, drag-and-drop file uploaders that accept financial PDFs. It displays a clear text list of supported file formats and shows processing indicators so the user knows the system is working.
- **Center Analytical Console:** This main section displays the visual graphs, including 3-Year Profit Trends and Loan-Reduction Charts. It also features a clear 'Risk Flag Warning Area' that flashes red if any payment anomalies are detected.
- **Right Interactive Assistant Panel:** This panel houses the conversational chat terminal. It comes complete with quick-click question suggestion chips, allowing users to rapidly explore data without needing technical skills.

4. Expected Business Benefits

- **Build Future Client Pipelines:** Instead of permanently losing applicants who are turned away, banks can offer an actionable checklist to fix their credit. This creates loyal future customers.
- **Prevent Loan Drop-Offs:** The system instantly suggests a smaller, alternative loan path. This ensures the bank does not lose the client entirely to competing institutions.
- **Optimize Revenue Generation:** The platform helps find and prioritize urgent operational applications, such as emergency payroll needs. This allows bank staff to close critical deals much faster.
- **Drastically Lower Processing Cycles:** Automating the reading of complex financial documents saves a massive amount of manual verification time for the credit team.
- **Insulate Bank Capital:** By testing borrower cash flows against future economic shifts, the system provides an extra safety shield. This keeps the bank's financial investments incredibly secure.

5. Simple Workflow

- 1. Upload Documents: The user uploads their financial files.
- 2. AI Reads Documents: The system processes the text and numbers.
- 3. Important Information is Retrieved: Key financial data points are extracted.
- 4. AI Creates Risk Summary & Actionable Recommendations: The platform generates an overview and suggests next steps.
- 5. Verification: A human reviews and signs off on the findings.
- 6. Final Result Displayed: The dashboard shows the final outcome.

6. Design Architecture and Technology Stack

How the System Works (Multi-Agent Architecture)

The platform runs on a modular, multi-agent structure designed for absolute accuracy with zero AI hallucinations. It operates in three main steps:

- 1. The Reading Step (Input Layer): This step takes the uploaded financial PDFs and securely breaks down text and numbers using automated parsing pipelines.
- 2. The Smart Brain (Orchestration Layer): An intelligent Orchestrator routes the data to specialized AI agents. A Retrieval Agent fetches document snippets, a Reasoning Agent computes risk scores, and a Recommender Agent builds credit-fix checklists. The system cross-checks answers to ensure absolute truthfulness.
- 3. The User Screen (Dashboard Layer): Finally, the system pushes the structured data directly into interactive trend charts, warning lights, and clickable chat widgets for the user.

Technology Stack Map (Using Azure Services)

Platform Component	Chosen Technology	Azure Service Mapping and Purpose
User Interface (Frontend)	Streamlit / React	Azure App Service: Hosts the visual dashboard and metrics charts.
Agent Orchestration	Python with LangGraph / LangChain	Azure Container Apps: Runs the multi-agent logic, state memory, and business rules.
Core Intelligence	Azure OpenAI Service	Provides enterprise-grade GPT-4o models, token embeddings,

		and strict data safety measures.
Vector Database	Azure AI Search	Manages vector and hybrid search with semantic ranking to ensure data retrieval without AI hallucinations.
State History & Memory	Azure Cosmos DB	Securely maintains persistent borrower chat logs, session variables, and system metadata.
Observability & Tracing	Azure Monitor + App Insights	Tracks backend log pipelines, token latency delays, overall system health, and request traces.

7. Project Administration, Repository, and Team Structure

GitHub Repository: <https://github.com/Chaurasiyakash0136/EY-Training-2026>

Standard Repository Architecture Plan

- **src/agents/ and src/orchestrator/:** These folders house the distinct core modules for the AI agents, including the logic scripts for Retrieval, Reasoning, and Recommendations.
- **src/ui/ and src/api/:** These directories contain the frontend Streamlit dashboards and the FastAPI backend endpoint routing interfaces.
- **infra/ and .github/workflows/:** These maintain the Azure Infrastructure as Code configurations. They also hold the automated CI/CD pipelines for linting and testing code.

Team Roles and Responsibilities

- **Team Lead / Project Manager (Scope Owner):** Manages overall development timeline tracking and leads daily 15-minute sync meetings. They also handle milestone mapping and resolve any execution roadblocks.
- **Solution Architect (Design Choice Owner):** Owns the formal multi-agent blueprint design. They manage framework integrations and establish system communication rules.

- **Agent / Backend Developer (Core Logic Builder):** Responsible for writing internal Python agent workflows. They hook up LangGraph state machines and structure the core prompt schemas.
- **Data / Vector Engineer (Pipeline Specialist):** Configures the parsing patterns used to read the uploaded financial PDFs. They also map the index settings inside Azure AI Search.
- **Frontend Developer (Interface Creator):** Builds the web dashboard layout, including the graphs, trend visualizers, and interactive user chat elements.
- **DevOps / Observability (Deployment & Quality Guard):** Sets up the Azure Container Apps pipelines and hooks up App Insights for tracing metrics. They also write the Test-Driven Development (TDD) test workflows.